BLOCKCHAIN EXPERIMENT - 9

AIM: Implement the Private Ethereum Blockchain using Geth.

THEORY:

Introduction to Ethereum

Ethereum is an open-source, decentralized blockchain platform that goes beyond simple cryptocurrency transactions by supporting programmable smart contracts and decentralized applications (DApps). It was proposed by Vitalik Buterin in 2013 and launched in 2015. Unlike Bitcoin which is primarily a digital currency, Ethereum functions as a programmable distributed computing platform where developers can deploy self-executing code called smart contracts. The native cryptocurrency of the Ethereum network is called Ether (ETH), which is used to pay for computational operations on the network, referred to as gas fees.

Ethereum maintains a global state that is updated with every transaction and smart contract execution. Every node in the network holds a copy of this state, ensuring decentralization and fault tolerance. The network uses an account-based model where each account has a balance and can interact with other accounts or smart contracts.

What is a Private Ethereum Network?

A private Ethereum network is a standalone blockchain that replicates the Ethereum protocol but operates in complete isolation from the public mainnet. Organizations, developers, and academic institutions use private networks to test and develop blockchain applications without incurring real transaction costs or exposing data to the public chain. All network parameters such as consensus mechanism, block time, gas limits, and initial account balances are fully customizable. This makes private networks ideal for controlled experimentation, proof-of-concept development, and educational purposes.

What is Geth (Go-Ethereum)?

Geth, short for Go-Ethereum, is the most widely used official implementation of the Ethereum protocol, written in the Go programming language. It allows a computer to function as a full Ethereum node capable of mining Ether, validating transactions, executing smart contracts, and interacting with the blockchain. Geth can be operated through a command-line interface, a built-in JavaScript console, or via JSON-RPC API calls. It supports both public Ethereum networks as well as custom private networks, making it the preferred tool for setting up and managing private Ethereum deployments.

Consensus Mechanism — Clique (Proof of Authority)

Private Ethereum networks typically use the Clique consensus algorithm instead of Proof of Work. Clique is a Proof of Authority (PoA) protocol where only a set of pre-approved accounts

called signers are authorized to produce and validate new blocks. Since block production does not involve solving computationally intensive puzzles, Clique is significantly faster and more energy efficient than traditional mining. The list of authorized signers is defined at the time of network creation and is embedded in the genesis block. This makes Clique well suited for private and consortium blockchain environments where participants are known and trusted.

Steps Involved in Implementing a Private Ethereum Network

1. Installing Geth: The first step involves downloading and installing the appropriate version of the Geth client on the system. Once installed, the binary is made accessible via the system PATH so it can be executed from any directory.

2. Creating Node Accounts: Each participating node requires an Ethereum account which consists of a public address and a corresponding private key stored in a keystore file. These accounts are created using the `geth account new` command and are protected by a password. The public addresses of these accounts are used in the genesis configuration.

3. Setting Up Password Files: To allow nodes to automatically unlock their accounts during startup without manual password entry each time, password files are created containing the account passwords in plain text.

4. Creating the Genesis Block: The genesis block is configured through a `genesis.json` file. This file defines the chain ID, consensus settings (Clique period and epoch), gas limit, initial Ether balances for each account, and the extradata field which encodes the authorized signer address. Every node in the network must be initialized with the same genesis file.

5. Initializing the Nodes: Each node's data directory is initialized using the `geth init` command along with the genesis file. This sets up the initial blockchain state and prepares the node to begin operating on the private network.

6. Running the Nodes: Node 1 is started as the signer node with mining enabled, responsible for sealing and producing new blocks. Node 2 is started as a member node that participates in the network but does not produce blocks. Both nodes are started with the Geth JavaScript console enabled for direct interaction.

7. Connecting the Nodes: Once both nodes are running, they are connected to each other using enode URLs. An enode is a unique identifier for a Geth node containing its public key and network address. Node 1's enode is retrieved using `admin.nodeInfo.enode` and then added to Node 2 using `admin.addPeer()`. Successful peering can be verified using `admin.peers`.

8. Sending Transactions: After the network is established and nodes are connected, Ether can be transferred between accounts using `eth.sendTransaction()` in the Geth console. The transaction is picked up by the signer node, included in a block, and confirmed on the private chain. Account balances can be verified using `eth.getBalance()`.

PROCEDURE:

Step 1: Install Geth

wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g2022_prajjwal_pandey@cloudshell:~$ wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version
--2026-04-09 15:28:07--  https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28248613 (27M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz'

geth-linux-amd64-1.13.14-2bd6bd01.tar.g 100%[===================================================================================>]  26.94M   765KB/s    in 39s

2026-04-09 15:28:47 (715 KB/s) - 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz' saved [28248613/28248613]

geth-linux-amd64-1.13.14-2bd6bd01/
geth-linux-amd64-1.13.14-2bd6bd01/COPYING
geth-linux-amd64-1.13.14-2bd6bd01/geth
Geth
Version: 1.13.14-stable
Git Commit: 2bd6bd01d2e8561dd7fc21b631f4a34ac16627a1
Git Commit Date: 20240227
Architecture: amd64
Go Version: go1.21.6
Operating System: linux
GOPATH=/home/g2022_prajjwal_pandey/gopath:/google/gopath
GOROOT=
g2022_prajjwal_pandey@cloudshell:~$
```

Step 2: Create folders and accounts for both nodes

```
g2022_prajjwal_pandey@cloudshell:~$ mkdir -p ~/eth-private/node1 ~/eth-private/node2
g2022_prajjwal_pandey@cloudshell:~$ geth113 --datadir ~/eth-private/node1 account new
INFO [04-09|15:36:17.038] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:36:17.041] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x93B0C547De2cF8228981Dd4a29aC3D0cA9a08859
Path of the secret key file: /home/g2022_prajjwal_pandey/eth-private/node1/keystore/UTC--2026-04-09T15-36-35.166702641Z--93b0c547de2cf8228981dd4a29ac3d0ca9a0885
9

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

g2022_prajjwal_pandey@cloudshell:~$
```

```
g2022_prajjwal_pandey@cloudshell:~$ geth113 --datadir ~/eth-private/node2 account new
INFO [04-09|15:38:49.547] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:38:49.549] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x6406586b898212f380A55Bbf5e6596bD95825D3b
Path of the secret key file: /home/g2022_prajjwal_pandey/eth-private/node2/keystore/UTC--2026-04-09T15-38-54.774381074Z--6406586b898212f380a55bbf5e6596bd95825d3
b

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Step 3: Create password files

```
g2022_prajjwal_pandey@cloudshell:~$ echo "Prajjwal" > ~/eth-private/node1/password.txt
echo "Prajjwal" > ~/eth-private/node2/password.txt
g2022_prajjwal_pandey@cloudshell:~$
```

Step 4: Create genesis.json and verify it looks correct

```
NODE1="PASTE_NODE1_ADDRESS_WITHOUT_0x_HERE"
NODE2="PASTE_NODE2_ADDRESS_WITHOUT_0x_HERE"
cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata":
"0x0000000000000000000000000000000000000000000000000000000000000000${NODE1}0
000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x${NODE1}": { "balance": "10000000000000000000" },
    "0x${NODE2}": { "balance": "10000000000000000000" }
  }
}
EOF
```

```
g2022_prajjwal_pandey@cloudshell:~$ NODE1="93B0C547De2cF8228981Dd4a29aC3D0cA9a08859"
NODE2="6406586b898212f380A55Bbf5e6596bD95825D3b"

cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000${NODE1}0000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x${NODE1}": { "balance": "100000000000000000000" },
    "0x${NODE2}": { "balance": "100000000000000000000" }
  }
}
EOF
g2022_prajjwal_pandey@cloudshell:~$
```

```
g2022_prajjwal_pandey@cloudshell:~$ cat ~/eth-private/genesis.json
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000093B0C547De2cF8228981Dd4a29aC3D0cA9a0885900000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x93B0C547De2cF8228981Dd4a29aC3D0cA9a08859": { "balance": "10000000000000000000" },
    "0x6406586b898212f380A55Bbf5e6596bD95825D3b": { "balance": "10000000000000000000" }
  }
}
g2022_prajjwal_pandey@cloudshell:~$
```

Step 5: Initialize both nodes

```
g2022_prajjwal_pandey@cloudshell:~$ geth113 --datadir ~/eth-private/node1 init ~/eth-private/genesis.json
INFO [04-09|15:49:20.956] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:49:20.959] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-09|15:49:20.963] Set global gas cap                       cap=50,000,000
INFO [04-09|15:49:20.963] Initializing the KZG library             backend=gokzg
INFO [04-09|15:49:21.035] Defaulting to pebble as the backing database
INFO [04-09|15:49:21.035] Allocated cache and file handles         database=/home/g2022_prajjwal_pandey/eth-private/node1/geth/chaindata cache=16.00MiB handles=
16
INFO [04-09|15:49:21.104] Opened ancient database                  database=/home/g2022_prajjwal_pandey/eth-private/node1/geth/chaindata/ancient/chain readonly=
false
INFO [04-09|15:49:21.105] State schema set to default              scheme=hash
INFO [04-09|15:49:21.105] Writing custom genesis block
INFO [04-09|15:49:21.109] Persisted trie from memory database      nodes=3 size=411.00B time=3.93038ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.0
0B
INFO [04-09|15:49:21.152] Successfully wrote genesis state         database=chaindata hash=e03c67..887468
INFO [04-09|15:49:21.152] Defaulting to pebble as the backing database
INFO [04-09|15:49:21.152] Allocated cache and file handles         database=/home/g2022_prajjwal_pandey/eth-private/node1/geth/lightchaindata cache=16.00MiB han
dles=16
INFO [04-09|15:49:21.219] Opened ancient database                  database=/home/g2022_prajjwal_pandey/eth-private/node1/geth/lightchaindata/ancient/chain read
only=false
INFO [04-09|15:49:21.219] State schema set to default              scheme=hash
INFO [04-09|15:49:21.219] Writing custom genesis block
INFO [04-09|15:49:21.223] Persisted trie from memory database      nodes=3 size=411.00B time=4.12979ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.0
0B
INFO [04-09|15:49:21.266] Successfully wrote genesis state         database=lightchaindata hash=e03c67..887468
g2022_prajjwal_pandey@cloudshell:~$
```

```
g2022_prajjwal_pandey@cloudshell:~$ geth113 --datadir ~/eth-private/node2 init ~/eth-private/genesis.json
INFO [04-09|15:49:52.109] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:49:52.111] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-09|15:49:52.115] Set global gas cap                       cap=50,000,000
INFO [04-09|15:49:52.116] Initializing the KZG library             backend=gokzg
INFO [04-09|15:49:52.189] Defaulting to pebble as the backing database
INFO [04-09|15:49:52.189] Allocated cache and file handles         database=/home/g2022_prajjwal_pandey/eth-private/node2/geth/chaindata cache=16.00MiB handles=
16
INFO [04-09|15:49:52.257] Opened ancient database                  database=/home/g2022_prajjwal_pandey/eth-private/node2/geth/chaindata/ancient/chain readonly=
false
INFO [04-09|15:49:52.257] State schema set to default              scheme=hash
INFO [04-09|15:49:52.257] Writing custom genesis block
INFO [04-09|15:49:52.262] Persisted trie from memory database      nodes=3 size=411.00B time=4.507603ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.
00B
INFO [04-09|15:49:52.306] Successfully wrote genesis state         database=chaindata hash=e03c67..887468
INFO [04-09|15:49:52.306] Defaulting to pebble as the backing database
INFO [04-09|15:49:52.306] Allocated cache and file handles         database=/home/g2022_prajjwal_pandey/eth-private/node2/geth/lightchaindata cache=16.00MiB han
dles=16
INFO [04-09|15:49:52.373] Opened ancient database                  database=/home/g2022_prajjwal_pandey/eth-private/node2/geth/lightchaindata/ancient/chain read
only=false
INFO [04-09|15:49:52.373] State schema set to default              scheme=hash
INFO [04-09|15:49:52.373] Writing custom genesis block
INFO [04-09|15:49:52.378] Persisted trie from memory database      nodes=3 size=411.00B time=4.105657ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.
00B
INFO [04-09|15:49:52.420] Successfully wrote genesis state         database=lightchaindata hash=e03c67..887468
g2022_prajjwal_pandey@cloudshell:~$
```

Step 6: Run Node 1 (signer)

```
cloudshell ×  cloudshell ×  +
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g2022_prajjwal_pandey@cloudshell:~$ export PATH="$HOME/bin:$PATH"
NODE1="93B0C547De2cF8228981Dd4a29aC3D0cA9a08859"
g2022_prajjwal_pandey@cloudshell:~$
```

```
g2022_prajjwal_pandey@cloudshell:~$ geth113 --datadir ~/eth-private/node1 \
  --networkid 1234 \
  --port 30303 \
  --http --http.port 8545 \
  --unlock 0x${NODE1} \
  --password ~/eth-private/node1/password.txt \
  --mine \
  --miner.etherbase 0x${NODE1} \
  --allow-insecure-unlock \
  console
INFO [04-09|15:52:04.688] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:52:04.691] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-09|15:52:04.694] Set global gas cap                       cap=50,000,000
INFO [04-09|15:52:04.695] Initializing the KZG library             backend=gokzg
INFO [04-09|15:52:04.767] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-09|15:52:04.767] Using pebble as the backing database
INFO [04-09|15:52:04.767] Allocated cache and file handles         database=/home/g2022_prajjwal_pandey/eth-private/node1/geth/chaindata cache=512.00MiB handles
=524,288
INFO [04-09|15:52:04.797] Opened ancient database                  database=/home/g2022_prajjwal_pandey/eth-private/node1/geth/chaindata/ancient/chain readonly=
false
INFO [04-09|15:52:04.798] State scheme set to already existing     scheme=hash
INFO [04-09|15:52:04.799] Initialising Ethereum protocol           network=1234 dbversion=<nil>
INFO [04-09|15:52:04.803]
INFO [04-09|15:52:04.803] ------------------------------------------------------------------------------------------------------------------------------------
-------------------
INFO [04-09|15:52:04.803] Chain ID:  1234 (unknown)
INFO [04-09|15:52:04.803] Consensus: Clique (proof-of-authority)
INFO [04-09|15:52:04.803]
INFO [04-09|15:52:04.803] Pre-Merge hard forks (block based):
INFO [04-09|15:52:04.803]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/h
omestead.md)
INFO [04-09|15:52:04.803]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/t
angerine-whistle.md)
INFO [04-09|15:52:04.803]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/s
purious-dragon.md)
INFO [04-09|15:52:04.803]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/s
```

```
INFO [04-09|15:52:04.971] Starting peer-to-peer node               instance=Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
INFO [04-09|15:52:04.991] New local node record                    seq=1,775,749,924,991 id=a9a288d6769625c8 ip=127.0.0.1 udp=30303 tcp=30303
INFO [04-09|15:52:04.994] IPC endpoint opened                      url=/home/g2022_prajjwal_pandey/eth-private/node1/geth.ipc
INFO [04-09|15:52:04.994] Started P2P networking                   self=enode://acfe33406144ee52760abb42a72a912f3123188c228f201424f105b391186998f0fc78a77e454680
d8898b9ca388cf647de953ae695bb21251e5f610baa5bb7d@127.0.0.1:30303
INFO [04-09|15:52:04.994] Generated JWT secret                     path=/home/g2022_prajjwal_pandey/eth-private/node1/geth/jwtsecret
INFO [04-09|15:52:04.995] HTTP server started                      endpoint=127.0.0.1:8545 auth=false prefix= cors= vhosts=localhost
INFO [04-09|15:52:04.996] WebSocket enabled                        url=ws://127.0.0.1:8551
INFO [04-09|15:52:04.996] HTTP server started                      endpoint=127.0.0.1:8551 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-09|15:52:06.133] Unlocked account                         address=0x93B0C547De2cF8228981Dd4a29aC3D0cA9a08859
INFO [04-09|15:52:06.133] Legacy pool tip threshold updated        tip=0
INFO [04-09|15:52:06.133] Legacy pool tip threshold updated        tip=1,000,000,000
INFO [04-09|15:52:06.133] Snap sync complete, auto disabling
INFO [04-09|15:52:06.134] Commit new sealing work                  number=1 sealhash=16e9bf..5ee3c4 txs=0 gas=0 fees=0 elapsed="200.259µs"
INFO [04-09|15:52:06.144] Successfully sealed new block            number=1 sealhash=16e9bf..5ee3c4 hash=6baf2d..432713 elapsed=10.718ms
INFO [04-09|15:52:06.145] Commit new sealing work                  number=2 sealhash=95fbef..db0a46 txs=0 gas=0 fees=0 elapsed="583.298µs"
INFO [04-09|15:52:06.149] Indexed transactions                     blocks=2 txs=0 tail=0 elapsed=3.670ms
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
coinbase: 0x93b0c547de2cf8228981dd4a29ac3d0ca9a08859
at block: 1 (Thu Apr 09 2026 15:52:06 GMT+0000 (UTC))
 datadir: /home/g2022_prajjwal_pandey/eth-private/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-09|15:52:07.394] New local node record                    seq=1,775,749,924,992 id=a9a288d6769625c8 ip=34.126.99.110 udp=30303 tcp=30303
INFO [04-09|15:52:11.010] Successfully sealed new block            number=2 sealhash=95fbef..db0a46 hash=4ad89a..cdb782 elapsed=4.864s
INFO [04-09|15:52:11.010] Commit new sealing work                  number=3 sealhash=6684cc..c8ae91 txs=0 gas=0 fees=0 elapsed="585.63µs"
INFO [04-09|15:52:15.529] Looking for peers                        peercount=0 tried=83 static=0
INFO [04-09|15:52:16.010] Successfully sealed new block            number=3 sealhash=6684cc..c8ae91 hash=d3d30d..b6a37c elapsed=4.999s
INFO [04-09|15:52:16.011] Commit new sealing work                  number=4 sealhash=2e769c..084439 txs=0 gas=0 fees=0 elapsed="672.511µs"
INFO [04-09|15:52:21.010] Successfully sealed new block            number=4 sealhash=2e769c..084439 hash=f585e4..392e05 elapsed=4.998s
INFO [04-09|15:52:21.010] Commit new sealing work                  number=5 sealhash=998266..47aa05 txs=0 gas=0 fees=0 elapsed="619.955µs"
INFO [04-09|15:52:25.681] Looking for peers                        peercount=0 tried=68 static=0
INFO [04-09|15:52:26.010] Successfully sealed new block            number=5 sealhash=998266..47aa05 hash=5865a2..bceb3e elapsed=4.999s
```

Step 7: Open new terminal tab and run Node 2 (member)

```
g2022_prajjwal_pandey@cloudshell:~$ export PATH="$HOME/bin:$PATH"
NODE2="6406586b898212f380A55Bbf5e6596bD95825D3b"
```

```
g2022_prajjwal_pandey@cloudshell:~$ geth113 --datadir ~/eth-private/node2   --networkid 1234   --port 30304   --http --http.port 8546   --unlock 0x${NODE2}   --
password ~/eth-private/node2/password.txt   --allow-insecure-unlock   console
INFO [04-09|16:10:33.186] Maximum peer count                       ETH=50 total=50
INFO [04-09|16:10:33.189] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-09|16:10:33.193] Set global gas cap                       cap=50,000,000
INFO [04-09|16:10:33.193] Initializing the KZG library             backend=gokzg
INFO [04-09|16:10:33.268] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-09|16:10:33.268] Using pebble as the backing database
INFO [04-09|16:10:33.268] Allocated cache and file handles         database=/home/g2022_prajjwal_pandey/eth-private/node2/geth/chaindata cache=512.00MiB handles
=524,288
INFO [04-09|16:10:33.300] Opened ancient database                  database=/home/g2022_prajjwal_pandey/eth-private/node2/geth/chaindata/ancient/chain readonly=
false
INFO [04-09|16:10:33.300] State scheme set to already existing     scheme=hash
INFO [04-09|16:10:33.301] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] ----------------------------------------------------------------------------------------------------------------------------------
------------------
INFO [04-09|16:10:33.302] Chain ID:  1234 (unknown)
INFO [04-09|16:10:33.302] Consensus: Clique (proof-of-authority)
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] Pre-Merge hard forks (block based):
INFO [04-09|16:10:33.302]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/h
omestead.md)
INFO [04-09|16:10:33.302]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/t
angerine-whistle.md)
INFO [04-09|16:10:33.302]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/s
purious-dragon.md)
INFO [04-09|16:10:33.302]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/s
purious-dragon.md)
INFO [04-09|16:10:33.302]  - Byzantium:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/b
yzantium.md)
INFO [04-09|16:10:33.302]  - Constantinople:               #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/c
onstantinople.md)
INFO [04-09|16:10:33.302]  - Petersburg:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/p
etersburg.md)
```

```
INFO [04-09|16:10:33.302] Post-Merge hard forks (timestamp based):
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] ------------------------------------------------------------------------------------------------------------------------------------
------------------
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] Loaded most recent local block           number=0 hash=e03c67..887468 td=1 age=57y1mo2d
WARN [04-09|16:10:33.303] Loaded snapshot journal                  diffs=missing
INFO [04-09|16:10:33.303] Initialized transaction indexer          range="last 2350000 blocks"
INFO [04-09|16:10:33.303] Loaded local transaction journal         transactions=0 dropped=0
INFO [04-09|16:10:33.325] Enabled snap sync                        head=0 hash=e03c67..887468
INFO [04-09|16:10:33.326] Gasprice oracle is ignoring threshold set threshold=2
WARN [04-09|16:10:33.330] Unclean shutdown detected                booted=2026-04-09T16:01:48+0000 age=8m45s
WARN [04-09|16:10:33.331] Engine API enabled                       protocol=eth
WARN [04-09|16:10:33.331] Engine API started but chain not configured for merge yet
INFO [04-09|16:10:33.331] Starting peer-to-peer node               instance=Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
INFO [04-09|16:10:33.363] New local node record                    seq=1,775,750,508,266 id=40d680c8eeed08a8 ip=127.0.0.1 udp=30304 tcp=30304
INFO [04-09|16:10:33.366] Started P2P networking                   self=enode://39855208b63767867bd73a27f0fee7a0a210588ad82c875d858df2c94a59e2d547e132ef6deda610
1cfba712be53477af7154b6ab8e6a03b814bf90fcdd60526@127.0.0.1:30304
INFO [04-09|16:10:33.366] IPC endpoint opened                      url=/home/g2022_prajjwal_pandey/eth-private/node2/geth.ipc
INFO [04-09|16:10:33.366] Loaded JWT secret file                   path=/home/g2022_prajjwal_pandey/eth-private/node2/geth/jwtsecret crc32=0x10258536
INFO [04-09|16:10:33.368] HTTP server started                      endpoint=127.0.0.1:8546 auth=false prefix= cors= vhosts=localhost
INFO [04-09|16:10:33.369] WebSocket enabled                        url=ws://127.0.0.1:8551
INFO [04-09|16:10:33.369] HTTP server started                      endpoint=127.0.0.1:8551 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-09|16:10:34.473] Unlocked account                         address=0x6406586b898212f380A55Bbf5e6596bD95825D3b
WARN [04-09|16:10:34.526] Served eth_coinbase                      reqid=3 duration="30.392µs" err="etherbase must be explicitly specified"
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
at block: 0 (Thu Jan 01 1970 00:00:00 GMT+0000 (UTC))
 datadir: /home/g2022_prajjwal_pandey/eth-private/node2
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-09|16:10:35.821] New local node record                    seq=1,775,750,508,267 id=40d680c8eeed08a8 ip=34.126.99.110 udp=30304 tcp=30304
INFO [04-09|16:10:43.413] Looking for peers                        peercount=0 tried=52 static=0
INFO [04-09|16:10:53.510] Looking for peers                        peercount=0 tried=29 static=0
```

Step 8: Connect the two nodes

In Node 1 console, get its enode

```
> admin.nodeInfo.enode
"enode://acfe33406144ee52760abb42a72a912f3123188c228f201424f105b391186998f0fc78a77e454680d8898b9ca388cf647de953ae695bb21251e5f610baa5bb7d@34.126.99.110:30303"
> INFO [04-09|16:22:47.013] Successfully sealed new block            number=231 sealhash=bb4346..610566 hash=d1fc70..84fdc7 elapsed=5.001s
INFO [04-09|16:22:47.014] Commit new sealing work                  number=232 sealhash=8aca46..08c3ca txs=0 gas=0 fees=0 elapsed="538.827µs"
INFO [04-09|16:22:52.010] Successfully sealed new block            number=232 sealhash=8aca46..08c3ca hash=d65637..b7846e elapsed=4.996s
INFO [04-09|16:22:52.010] Commit new sealing work                  number=233 sealhash=e1df14..1d6155 txs=0 gas=0 fees=0 elapsed="485.675µs"
INFO [04-09|16:22:52.270] Looking for peers                        peercount=0 tried=60 static=0
INFO [04-09|16:22:57.011] Successfully sealed new block            number=233 sealhash=e1df14..1d6155 hash=0be857..1fcf8d elapsed=5.001s
INFO [04-09|16:22:57.012] Commit new sealing work                  number=234 sealhash=51ca87..5b8c95 txs=0 gas=0 fees=0 elapsed="501.891µs"
```

Now, switch to Node 2 console and paste the Node 1 enode

```
> admin.addPeer("enode://acfe33406144ee52760abb42a72a912f3123188c228f201424f105b391186998f0fc78a77e454680d8898b9ca388cf647de953ae695bb21251e5f610baa5bb7d@34.126
.99.110:30303")
true
> INFO [04-09|16:25:38.021] Looking for peers                        peercount=0 tried=45 static=1
```

In Node 2 console, check they are connected

```
> admin.peers
[{
    caps: ["eth/68", "eth/69", "snap/1"],
    enode: "enode://00162513d2f3e10b7ccd0dc4a3d139842e3f9681147bf7367aa7fe40421ce3904f8a4ea0db1254cc2d0d8a0dfec08457c3938017bf424a952c92c70931183abb@47.187.149.99:31404",
    id: "26184e724beb8194376e59f1fd0493aee8d3cb40469857695c966147c8b8aeeb",
    name: "Nethermind/v1.36.0+31cb81b7/linux-x64/dotnet10.0.1",
    network: {
      inbound: false,
      localAddress: "10.88.0.4:55586",
      remoteAddress: "47.187.149.99:31404",
      static: false,
      trusted: false
    },
    protocols: {
      eth: "handshake",
      snap: "handshake"
    }
}]
```

Step 9: Send a transaction

In Node 1 console, send transaction to Node 2

```
> eth.sendTransaction({
......   from: eth.accounts[0],
......   to: "0x5fd63344c1f40712A9A85d9d7e2B304a6C558716",
......   value: web3.toWei(1, "ether")
...... })
INFO [04-05|13:43:48.276] Setting new local account                address=0x1D8b6d4F6C677D2fB66CA97FbD81bED2cD23c4D3
INFO [04-05|13:43:48.276] Submitted transaction                    hash=0x566ac1b0f280ebd7b6d4a6899141ba3624 8ed3ec43bd24f8a08af6badab79089 from=0x1D8b6d4F6C677D2fB66CA97FbD81b
ED2cD23c4D3 nonce=0 recipient=0x5fd63344c1f40712A9A85d9d7e2B304a6C558716 value=1,000,000,000,000,000,000
"0x566ac1b0f280ebd7b6d4a6899141ba3624 8ed3ec43bd24f8a08af6badab79089"
```

and check the balance of Node 2

```
> eth.getBalance("0x5fd63344c1f40712A9A85d9d7e2B304a6C558716")
10100000000000000000
```

CONCLUSION

This experiment successfully demonstrated the implementation of a private Ethereum blockchain network using the Geth (Go-Ethereum) client. A two-node private network was set up from scratch, beginning with the installation of Geth, creation of individual node accounts, and configuration of the genesis block using the Clique Proof of Authority consensus mechanism. Both nodes were initialized with the same genesis file, started with appropriate network parameters, and successfully connected to each other using enode URLs.

Once the network was operational, the signer node (Node 1) began producing and sealing blocks at regular intervals, confirming that the consensus mechanism was functioning correctly. The two nodes were peered together and their connection was verified using the `admin.peers` command. A transaction was then executed from Node 1 to Node 2 using `eth.sendTransaction()`, and the updated balance of Node 2 was confirmed using `eth.getBalance()`, validating that transactions are being processed and recorded on the private chain.

Through this experiment, a practical understanding of how Ethereum nodes operate, how private blockchain networks are configured, and how peer-to-peer communication works between nodes was gained. It also highlighted the advantages of using Proof of Authority consensus in private networks, particularly in terms of faster block production and reduced computational overhead compared to Proof of Work. Overall, this experiment provided hands-on exposure to core blockchain concepts including genesis block configuration, node initialization, peer discovery, and on-chain transaction verification.